

Last Time:

□ Intro to HRL

This Time:

□ sequential decision-making

□ MDPs

□ (Policies, Values, Bellman Eqn.)

Lecture 2

HRL, FALL '26

Andrea Bajcsy

What is sequential decision-making?

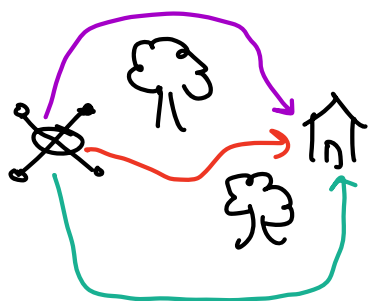
Sequential decision-making involves making a sequence of decisions over time, where each decision affects future outcomes and decisions.

↳ different from one-shot decisions which are self-contained single decisions (e.g. img. classification)

Why do we care?

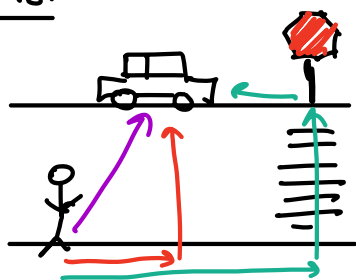
Sequential decision-making is everywhere - play games, planning a trip or career - and its present in interaction!

In RL, sequential decision-making will form the "mathematical backbone" of how we model people, robots, and their interactions.



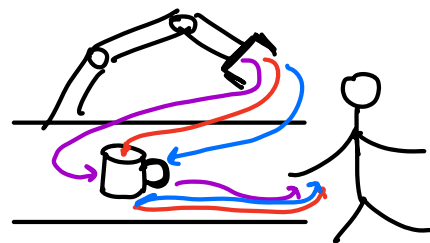
robot must plan ahead considering long-term consequences

⇒ How will R fly home?



humans as sequential decision-makers: we need to model them well!

⇒ How will H go to their car?



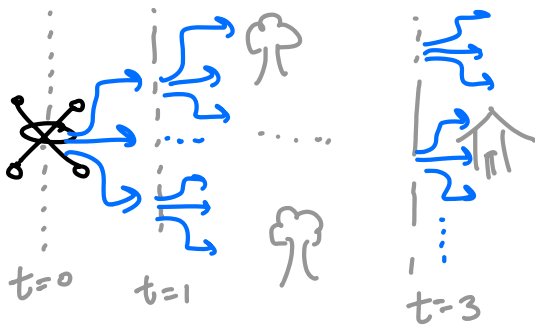
interaction as an ongoing exchange where the agents influence each other over time.

⇒ How should R hand H the mug?

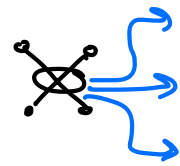
Why is it hard?

[A1] decision-tree grows exponentially in time horizon.
↳ too many possibilities to enumerate / compute

↳ * also grows exponentially in num. agents!

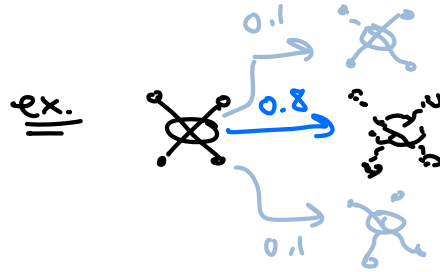
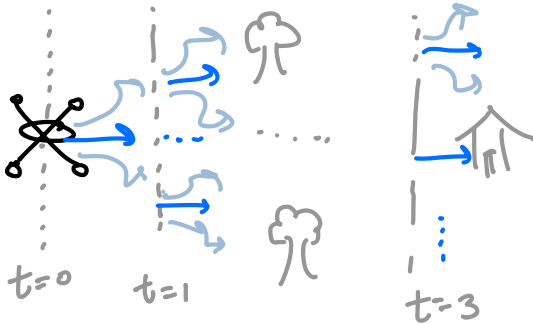


There are $|A|^{T+1}$
possible sequences
of decisions
 $= 3^4 = 81$ seq.



$$|A| = 3$$

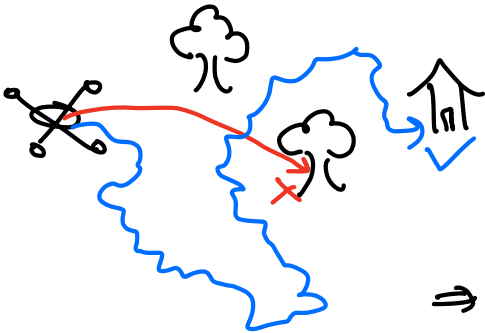
A2 outcome of taking actions can be **stochastic!**
 $\hookrightarrow$ the same action doesn't always have the same result.



ex. even if
R chooses to go
straight, wind
pushes left/right
sometimes.

Even w/ the same action sequence you
planned, the outcomes might not be same!

A3 Hard to **assign credit** to the right past action(s)



✓ successfully made it home, but
was it all good..?

X fail to get home, but was it all bad...?

$\Rightarrow$ if rewards come late, hard to tell which
actions were good / bad!

Markov Decision Processes (MDPs)

MDPs are a mathematical model for sequential decision-making
in a fully observable, stochastic, discrete-time environment
with a Markovian transition model.

$$M = \langle S, A, T, r \rangle$$

- states: $s \in S \rightarrow$ state space

- actions: $a \in A$ → action space

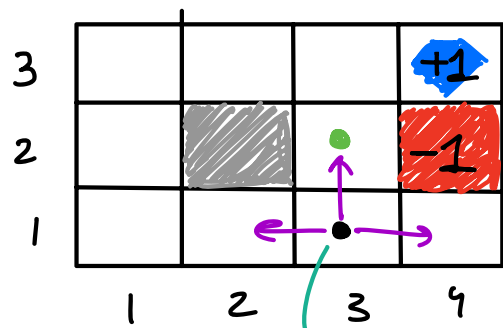
- transition function: $T: S \times A \rightarrow S$

$$T(s, a, s') \quad \text{or}$$

$$P(s' | s, a) \quad \text{or}$$

$s' \sim P(\cdot | s, a)$
is sampled from

$$P(s' = (2, 3) | s = (1, 3), a = \uparrow) = 0.8$$



$A = \{\leftarrow, \uparrow, \rightarrow\}$

$s = (1, 3)$
row col

- reward function: $r: S \times A \times S \rightarrow \mathbb{R}$

$$\text{or } r(s, a, s') \quad \text{or } r(s')$$

$$r(s = (2, 3), a = \rightarrow, s' = (2, 4)) = -1$$

Ⓚ What is "Markov" about this?

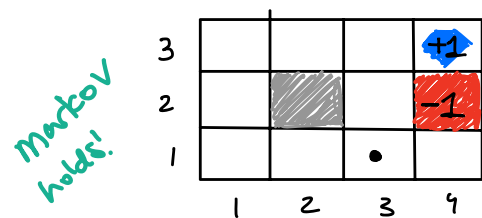
The transition function is where the Markov part comes into play. Markov Property: the future is independent of the past, given the present.

⇒ This means that the action outcomes depend only on the current state, and not the history!

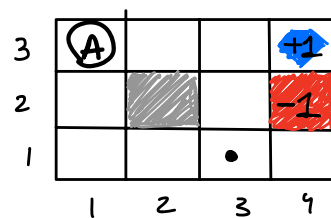


! makes planning / solving MDPs easier!

$$P(s_{t+1} | s_t, a_t, s_{t-1}, a_{t-1}, \dots, s_0) = P(s_{t+1} | s_t, a_t)$$



$s = (x, y)$



Markov doesn't hold!

"Get the gem" →

"Get the gem after visiting (A)"

! If a problem is Markov depends on how we define the state + task.

⇒ But, if $s = (x, y, A_visited)$ then yes!

Policies

The object we ultimately want to solve for; or knowing what is the best decision to make @ any state.

$\pi: S \rightarrow A$
deterministic $\pi(s)$

$\pi: S \rightarrow \Delta A$
stochastic $\pi(a|s)$ probability simplex
"distribution over actions"

ex. policy $\rightarrow$

$\pi: S \rightarrow A$

3	$\rightarrow$	$\rightarrow$	$\rightarrow$	$+1$
2	$\rightarrow$		$\rightarrow$	-1
1	$\rightarrow$	$\rightarrow$	$\rightarrow$	$\rightarrow$
	1	2	3	4

Evaluate the quality of a policy π by its expected, cumulative reward $\otimes$ we will get to expected bit in a second...

① "roll out" / simulate π from each $s_0 \in S$ to get a bunch of trajectories

$$\{ s_0, \underset{a_0}{\pi(s_0)}, s_1, \underset{a_1}{\pi(s_1)} \dots \}$$

② Evaluate the utility of the rollout; cumulative reward

$$R(s_0, s_1, s_2 \dots) \triangleq r(s_0) + r(s_1) + r(s_2) + \dots = \sum_{t=0}^{\infty} r(s_t)$$

$\pi_A: S \rightarrow A$

3	$\rightarrow$	$\rightarrow$	$\rightarrow$	$+1$
2	$\rightarrow$		$\rightarrow$	-1
1	$\rightarrow$	$\rightarrow$	$\rightarrow$	$\rightarrow$
	1	2	3	4

$\pi_B: S \rightarrow A$

3	$\rightarrow$	$\rightarrow$	$\rightarrow$	$+1$
2	$\uparrow$		$\uparrow$	-1
1	$\uparrow$	$\rightarrow$	$\uparrow$	$\leftarrow$
	1	2	3	4

better policy
b/c more
R!

optimal policy ($\pi^*: S \rightarrow A$) yields the highest cumulative reward.

! What counts as good depends on reward!

Sparse vs. Dense Rewards

Problem: If I only reward +1 at the goal (-1 @ loss)
there are many policies which look equally good.

⊗ Underspecified rewards are a huge problem in AI / HRI

Discount Factor

- denoted by $\gamma \in [0, 1] \Rightarrow$ sometimes: $\text{MDP} = \langle S, A, T, r, \gamma \rangle$
- describes an agent's preference for current rewards over future ones.
 $\gamma < 1 \Rightarrow$ more short-term, $\gamma = 1 \Rightarrow$ longer term

$$R(s_0, s_1, s_2, \dots) = \gamma^0 r(s_0) + \gamma^1 r(s_1) + \gamma^2 r(s_2) \dots = \sum_{t=0}^{\infty} \gamma^t r(s_t)$$

$\Rightarrow$ discounted cumulative reward $\rightarrow$

$\rightarrow$ connections to psych + econ

⊗ "Models of Temporal Discounting 1937 - 2000" by Till Grüne-Yanoff.
discounting can be a good model of human + animal behavior

Ultimately, we are searching for a policy $\pi^*: S \rightarrow A$ that
maximizes the expected cumulative discounted reward:

$$\pi^* = \arg \max_{\pi} \mathbb{E}_{\tau \sim p_{\pi}(\tau)} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \right]$$

return the policy which maximizes... $\tau := (s_0, a_0, s_1, a_1, \dots)$

rollouts are stochastic! So take expectation over all trajectories induced by policy π

$$p_{\pi}(\tau) = p(s_0) \prod_{t=0}^{\infty} p(s_{t+1} | s_t, a_t)$$